

Article

Design and Preparation of PU/EP Blend Resin Grafted by Hydrophilic Molecular Segments

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Abstract: Film-forming resins, as carriers of antifouling coatings, play an essential role in the functional performance of coatings. With the gradual promotion of coatings in the field of marine antifouling, the preparation of film-forming resins containing hydrophilic chain segments is urgently needed to give full play to the functional and time-sensitive performance of antifouling coatings. In this paper, the single-ended isocyanate group (NCO) polyurethane (PU) prepolymer with hydrophilic chain segments was prepared and grafted onto an epoxy resin (EP) molecular chain to obtain the PU/EP grafted blend. Successful preparation of the PU/EP grafted blends was confirmed by infrared spectrum analysis. Simultaneously, the grafted blend was cured, and its hydrophilicity and mechanical properties were also tested. The results demonstrate that the grafted method can effectively improve the hydrophilicity of the film-forming resin while maintaining its mechanical properties. It can be concluded from the hydrophilic and mechanical properties of the materials that the comprehensive properties of PU/EP grafted blends are the best when the PU content is between 40% and 50%. The hanging plate test showed that the antifouling effect of the antifouling coating prepared by hydrophilic modified PU/EP film-forming resin was better than that of the traditional antifouling coating under the experimental conditions.

Keywords: film-forming resin; antifouling coating; hydrophilicity; polyurethane (PU); epoxy resin (EP); grafted blend; mechanical property



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1. Introduction

Antifouling coating is a kind of functional coating that is frequently applied to the surfaces of marine installations, such as ships, to block or delay the attachment of marine fouling organisms [1]. In this way, the wastage of power energy and human resources [2–4] can be significantly reduced. Antifouling coating is composed of film-forming resin and additives, such as an antifouling agent and coupling agent, each with its function [5]. At present, most of the research on antifouling coatings covers two perspectives: improving the application environment matching of film-forming resin and investigations on the performance of antifouling agents and other additives. The latter can cause fatal damage to marine fouling organisms. However, the contradiction between toxic pollution and marine ecological maintenance cannot be handled well in a short time since the development of a nontoxic or low toxic antifouling agent that completely replaces toxic antifouling agents and complies with the law will undergo the process of screening, elimination and re-screening [5,6]. Moreover, film-forming resin is the carrier of antifouling agents and other added materials. Therefore, the functional film-forming resin must be researched in the future to improve the matching of its application environment as much as possible and thus improve the effect of antifouling coatings, though this method is slightly conservative compared with the development of a new antifouling agent [6]. The performance of film-forming resin, which functions as the dispersing carrier of functional additives, directly

determines the functionality and timeliness of antifouling coatings. Hence, it is necessary to develop high-performance film-forming resin that is suitable for specific applications [7].

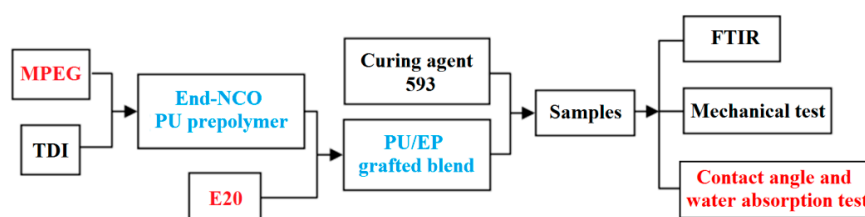
Considering that it is easy for ship coatings to peel off under the shear stress of water flow at high speeds, excellent film-forming resin mechanical properties [8] are required, because the mechanical properties of the coatings are mainly associated with film-forming resin. At present, film-forming resins are primarily composed of EP and PU. However, there are still many limitations to their performance. For example, EP has poor toughness and impact resistance [9]. PU has the advantages of high elasticity, good impact resistance, and molecular design, while its mechanical strength is insufficient compared with that of EP. Therefore, epoxies and polyurethanes provide the best overall combination of film properties compared with other organic coatings [10]. The two materials are physically blended to prepare film-forming resin for coatings, contributing to maximization of the performance advantages of the two materials [11]. However, a poor hydrophilicity problem began to appear with the promotion of the above resin in the field of antifouling. This makes it difficult for antifouling agents to be released from the coating, leading to a remarkable decrease in the functional effect of the antifouling coating [12]. Besides, the hydrophilicity of coatings can positively affect the adhesion of protein fouling organisms, and its antiadhesion mechanism is a research hotspot [13–15]. Studies aimed at determining how to properly improve the hydrophilicity of materials without reducing the comprehensive performance of film-forming resin are becoming increasingly essential in the field of marine antifouling. There have been many cases of chemical modification of PU and EP. The effects of hydrogen bonding on the free volume and miscibility were first investigated for PU/EP interpenetrating polymer network (IPN) nanocomposites. The results demonstrated that stronger the hydrogen-bonding interactions are associated with a higher average chain packing efficiency, and smaller free volume hole sizes lead to better miscibility [16]. Studies have revealed that the PU/EP interpenetrating network structure prepared by the microwave curing method not only has the same structure as thermal curing but it also has a shortened curing time. The tensile properties of microwave cured IPN are better than those of thermal cured IPN, while the impact strength of thermal cured IPN is slightly higher [17]. Some studies have also suggested that after blending PU with EP, the damping performance of EP composites significantly increases with the increase in frequency, and the damping temperature range moves toward higher temperatures [18]. In recent years, the tribological properties of polyurethane/epoxy interpenetrating network (PU/EP-IPN) composites have been explored to guide the research and development of polymer friction materials under water lubrication. The results indicate that the friction and wear properties in the water-lubricated medium are dramatically improved by adding different kinds of fillers (SiC submicron particles and short carbon fibers (SCF)) [19].

As revealed from the above research, PU/EP materials have been widely studied, the technology is relatively mature, and the comprehensive properties have been optimized and improved [20]. These studies have inspired the preparation of PU/EP hydrophilic graft blends. In this paper, a hydrophilic PU molecular chain was designed and then grafted onto the EP to effectively improve the hydrophilicity of PU/EP grafted blends without affecting the comprehensive properties of the material. It is a potential method that could be used to improve the antifouling effect of new antifouling coatings.

2. Experiments

Firstly, PU prepolymer with single-ended NCO containing the hydrophilic polyethylene glycol (PEG) chain was prepared. Afterwards, hydrophilic PU/EP grafted blends were obtained by grafting NCO with a hydroxyl group in EP. Finally, an adduct of diethylenetriamine and butyl glycidyl ether (593 curing agent) was employed to cure and form the blends [21]. The experimental design is illustrated in Scheme 1.

Polyethylene glycol monomethyl ether (MPEG); 2, 4-toluene diisocyanate (TDI); polyurethane (PU); epoxy resin (EP: E20); adduct of diethylenetriamine and butyl glycidyl ether (593 curing agent); Fourier Infrared Spectrometer (FTIR).



Scheme 1. Design process.

2.1. Experimental Reagents

The main reagents used in the experiment are presented in Table 1.

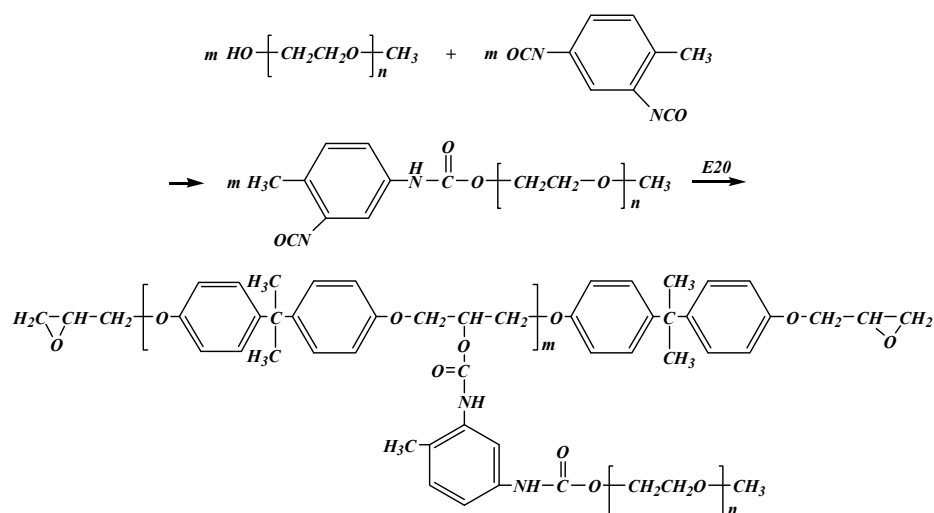
Table 1. Main experimental reagents.

Reagents	Purity	Manufacturer
N, N-Dimethylformamide (DMF)	AP	Tianjin Guangfu Chemical Co., Ltd., Tianjin, China
Butyl acetate	AR	Tianjin Guangfu Chemical Co., Ltd., Tianjin, China
2, 4-toluene diisocyanate (TDI)	CP	Haikeda Chemical Co., Ltd., Haikeda, China
Polyethylene glycol monomethyl ether (MPEG, the molecular weight is 1000)	AR	Nantong Haitianyuan Chemical Co., Ltd., Nantong, China
EP (E20)	AR	Zhengzhou Haoyuan chemical Co., Ltd., Zhengzhou, China
Di-n-butylamine	AR	Tianjin Bodi Chemical Co., Ltd., Tianjin, China
Hydrochloric acid (HCl)	AR	Haikeda Chemical Co., Ltd., Haikeda, China
Adduct of diethylenetriamine and butylglycidyl ether (593curing agent)	CP	Zhengzhou Haoyuan chemical Co., Ltd., Zhengzhou, China
Cu ₂ O	AR	Beilian Fine Chemical Co., Ltd., Beilian, China
Fe ₂ O ₃	AR	Tianjin kemio Co., Ltd., Tianjin, China
Air SiO ₂	CP	Haihua Co., Ltd., Weifang, China
KH550	CP	Xince Co., Ltd., Tianjin, China
Commercial antifouling coating	CP	Kalin Co., Ltd., Shenzhen, China
Marine anticorrosive paint	CP	Jiren Co., Ltd., Wuhan, China

2.2. Preparation of Hydrophilic PU/EP Grafted Blends

2.2.1. Principle of Synthesis

Single-ended NCO PU prepolymers were prepared by reacting MPEG with TDI in a molar ratio of 1:1. Then, PU/EP grafted blends were obtained by grafting NCO in the PU prepolymer with a hydroxyl group in the EP. The specific reaction formula used was (Scheme 2).



Scheme 2. Design route for the PU/EP grafted blends.

2.2.2. Detailed Synthesis Steps

First, E20-butyl acetate solution with a mass ratio of 1:1 was prepared. Then, under the protection of dry nitrogen, a certain amount of TDI was transferred to a three-mouth flask, to which the dehydration treated MPEG-1000 (molar ratio TDI/MPEG = 1.05/1) was added. The reaction took 6 h and was conducted at 70 °C. The NCO content in the system was determined by the Di-n-butylamine method. The PU prepolymer containing single-ended NCO was obtained when the NCO content reached the theoretical value. Next, the PU prepolymer was dropped into the prepared E20 solution. (The mass ratio formula of PU and EP is shown in Table 2). The reaction time was 5 h and it was conducted at 70 °C. The PU/EP grafted blend was acquired when the intensity of the NCO absorption peak of the product disappeared. Meanwhile, the epoxy value of the blend was measured.

Note: the actual weighed amount of each component was calculated according to the amount required for each experimental test.

Table 2. The mass ratio formula of PU and EP.

PU	70	60	50	40	30
EP	30	40	50	60	70

2.3. Preparation of Solid Test Samples

A measured 593 curing agent was added to the PU/EP grafted blend. Then, the mixture was put into an oven for 24 h at 100 °C to prepare the samples.

3. Characterization and Testing

3.1. Characterization and Test Devices

Information about the instruments and equipment used for characterization and testing is presented in Table 3.

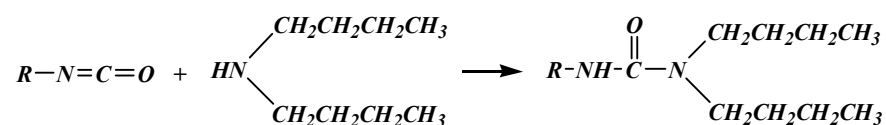
Table 3. Test and characterization devices and manufacturers.

Devices	Type	Manufacturer
Fourier Infrared Spectrometer (IR)	Spectrum BX II	Perkin Elmer, OH, USA
Dynamic Thermal Mechanical Analyzer (DMA)	METTLER TOLEDO	Mettler company, Greifensee, Switzerland
Electronic universal testing machine	TH-500N	JiangduTianhui Test Machinery Co., Ltd., Jiangdu, China
Contact Angle tester	JGW-360a	Chengde Shenghui Testing Machine Co., Ltd., Chengde, China

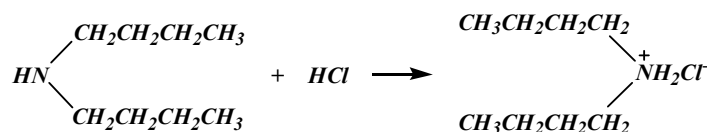
3.2. Characterization and Test Methods

3.2.1. Determination of the Mass Fraction of NCO in the PU Prepolymer

The NCO content in the prepolymer was determined by the di-n-butylamine method. As shown below (Schemes 3 and 4):



Scheme 3. Reaction equation for excess di-n-butylamine and NCO.



Scheme 4. Reaction equation for HCl solution and residual di-n-butylamine in the mixture.

First, 3 g of the prepolymer was weighed and put into a conical flask, to which 10 mL of 1 mol/L di-n-butylamine toluene solution was added, and they were mixed evenly. Next, 50 mL of anhydrous ethanol and a few drops of bromocresol green indicator were added to the conical flask and titrated with 0.5 mol/L hydrochloric acid solution. Meanwhile, the blank experiment was completed. Finally, the NCO mass percentage was calculated as follows:

$$\text{NCO}\% = \frac{4.202(V_b - V_s)}{m_s} c(\text{HCl}) \times 100\% \quad (1)$$

where m_s , V_b , V_s , and c represent the mass of the prepolymer (g), the titration volume (mL) in the blank test, the titrated volume (mL) of the sample, and the concentration of the standard HCl solution (mol/L), respectively.

3.2.2. Determination of the Epoxy Value

The epoxy value indicates the amount of material containing epoxy groups in the EP per 100 g. It can be determined by the acetone hydrochloride method.

Acetone hydrochloride solution was prepared by combining 4 mL of concentrated hydrochloric acid with 100 mL of acetone. Then, 3 g of the sample was put into a 250 mL conical flask, to which 25 mL of the prepared acetone hydrochloride solution was added, and the mixture was shaken well. Next, 25 mL of anhydrous ethanol and a few drops of bromocresol green indicator were added and titrated with 0.5 mol/L NaOH solution. Meanwhile, the blank experiment was finished. Finally, the epoxy value (EV) was calculated according to the following formula:

$$\text{EV} = \frac{(V_0 - V_1)}{10m} c(\text{NaOH}) \quad (2)$$

where m , V_0 , V_1 , and c represent the mass of the sample (g), the volume (mL) of the NaOH standard solution consumed in the control test, the volume (mL) of the NaOH standard solution consumed by the sample, and the concentration of the standard solution of NaOH (mol/L), respectively.

3.2.3. Infrared Spectroscopy Characterization

PU prepolymers and graft blends were characterized by the Spectrum BX II infrared spectrometer produced by PerkinElmer instruments, in which the liquid was coated with 32 scans in the wavenumber range of 400–4000 cm^{-1} and a resolution of better than 0.8 cm^{-1} .

3.2.4. Contact Angle Test

The JGW-360A contact angle tester was used for testing. Water drops were dropped onto the film surface in the form of hanging drops. The static water contact angle was determined by observing the angle between the water drops and the film surface with a tester. Three points on the diagonal of each sample surface were measured, and the average value was taken.

3.2.5. Water Absorption Test

The water absorption rate of the sample in the artificial seawater was tested under the standard GB/T 1034-1998. First, the sample was made into a standard rectangular piece with an area of $2 \text{ m}^{-2} \times 1 \text{ m}^{-2}$. Then, it was immersed in artificial seawater at 25 °C. The weight of the sample was taken before and after the immersion in parallel three times.

Finally, the water absorption rate of the sample at different times was calculated using the following formula.

$$Q_t = \frac{m_t - m_0}{m_0} \times 100\% \quad (3)$$

where Q_t , m_0 , and m_t represent the water absorption rate, the mass before water absorption (g), and the mass after water absorption at time t (g), respectively.

3.2.6. Tensile Property Test

Based on the standard GB/T 528-2009, the tensile rate of the electronic universal testing machine and the number of tensile samples were set at 500 mm/min and 7, respectively. Finally, the maximum and minimum values were removed from the data, and the average value was taken.

3.2.7. Shear Strength Test

According to the standard GB/T 7124-2008, the shear strength of the coating was tested by a universal testing machine. The number of samples was 7. Finally, the maximum and minimum values were removed from the data, and the average value was taken.

3.2.8. Dynamic Mechanical Performance Test

The dynamic mechanical properties of solid samples were tested with DMA₁. The test conditions were set as follows: frequency of 1 Hz, heating rate of 3 K/min, test temperature range of −80–100 °C, and a sample size of 50 mm × 6 mm × 2 mm.

3.2.9. Ocean Hanging Plate Test

According to the standard GB/T 7124-2008, the sample was immersed into seawater on the Hainan Hanging Plate Test Base and salvaged regularly to observe and record the adhesion of marine organisms on the sample surface.

4. Results and Discussion

4.1. FTIR Analysis of PU Prepolymers before and after Grafting

Figure 1 exhibits the infrared spectra of the two reaction products. The absorption peak of the amide bond appears at 1700 cm^{−1} in both curves. The single-ended NCO prepolymer presents an NCO bond absorption peak at 2250 cm^{−1}, in which the other curve disappears. This suggests that the NCO group in the latter reacts completely. Meanwhile, the latter has an epoxy bond absorption peak at 910 cm^{−1}, which is consistent with the results of the design route. The results suggest that the reaction product was the PU/EP grafted blend.

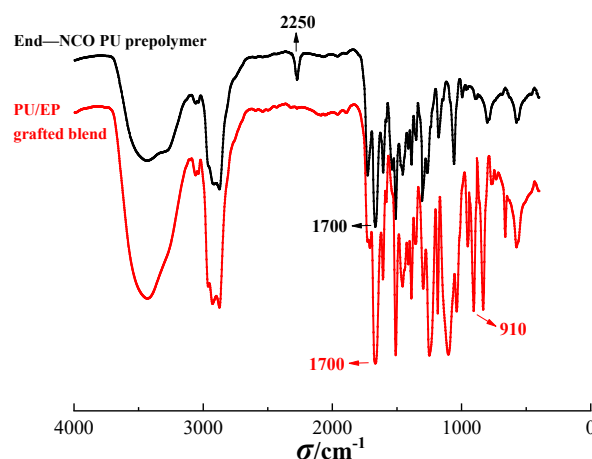


Figure 1. IR spectra of the PU prepolymer and EP before and after grafting. Polyurethane (PU); Epoxy resin (EP: E20).

4.2. Hydrophilicity of PU/EP Grafted Blends

Figure 2 presents the contact angle data for graft blends with different PU contents. As observed from the figure, the contact angle of the blend gradually decreased as the PU content increased. The contact angle of water decreased from 62.4 to 47.2 as the content of PU increased from 0.3 to 0.7. The above data reveal that the hydrophilicity of the material was improving and was positively affected by a high content of hydrophilic components, a greater PU content, and by having more hydrophilic molecular segments [22].

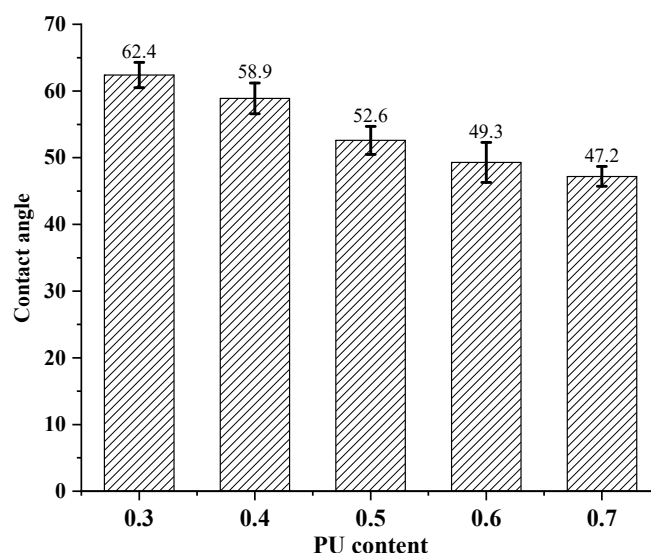


Figure 2. Water contact angles of PU/EP grafted blends.

4.3. Water Absorption Properties of PU/EP Grafted Blends

Figure 3 displays the relationship between water absorption and the soaking time of PU/EP grafted blends. Specifically, the water absorption of PU/EP grafted blends gradually increased as the PU content increased. The stable value of water absorption increased from 6% to 17% as the content of PU increased from 0.3 to 0.7. Meanwhile, the increasing range of water absorption was associated with the PU content due to the hydrophilic peg segment in the PU component. Besides, the water absorption of the sample became stable after 400 min. Different from the water-absorbent materials made of pulp, which absorb water through the capillary phenomenon, polyurethane absorbs water between chain segments using hydrophilic molecular chains until the water absorption performance between chain segments reaches saturation [23].

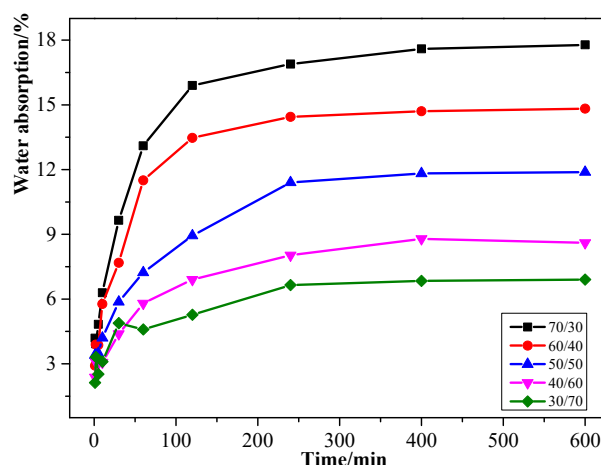


Figure 3. Water absorption of PU/EP grafted blends.

4.4. Mechanical Properties of PU/EP Grafted Blends

Figure 4 shows the relationship between the mechanical strength of PU/EP grafted blends and the mass fraction of PU. With an increase in PU content, the tensile strength of PU/EP grafted blends increases, while the shear strength decreases to a certain extent. Although the toughness of EP is insufficient, its strength is high compared with that of PU [9,11]. Therefore, the strength of grafted blends is affected by a decrease in the EP content. From another perspective, the PU content is related to the adhesive strength of the material. Thus, the shear strength of PU/EP grafted blends is improved with an increase in the PU content. Therefore, it is best to control the PU content in PU/EP grafted blends to be not more than 50% to fulfil considerations of both tensile strength and shear strength.

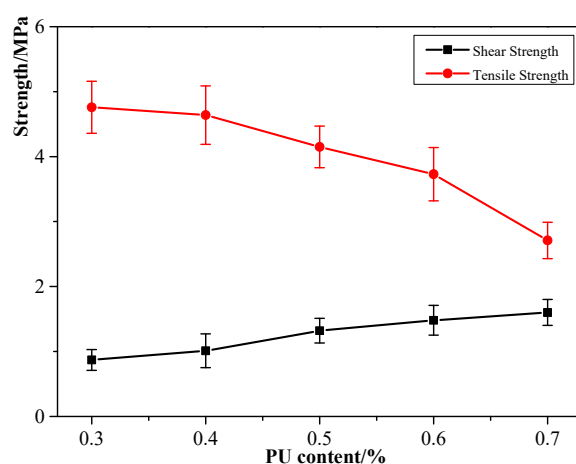


Figure 4. Mechanical properties of PU/EP grafted blends.

4.5. Dynamic Mechanical Properties of PU/EP Grafted Blends

Figure 5 provides the dynamic mechanical spectra of PU/EP grafted blends with different PU contents. Table 4 presents the corresponding dynamic mechanical data. These results show that the glass transition temperature of PU/EP grafted blends gradually increases with a decrease in the PU content. This is because the glass transition temperature of EP is very high, and the glass transition temperature of grafted blends moves to higher temperatures when the content of EP increases. In particular, the lower the PU content is, the wider the width of the damping peak is, suggesting worse compatibility between PU and EP. According to the compatibility shown between PU and EP in the dynamic mechanical property diagram, the PU content should not be less than 40%.

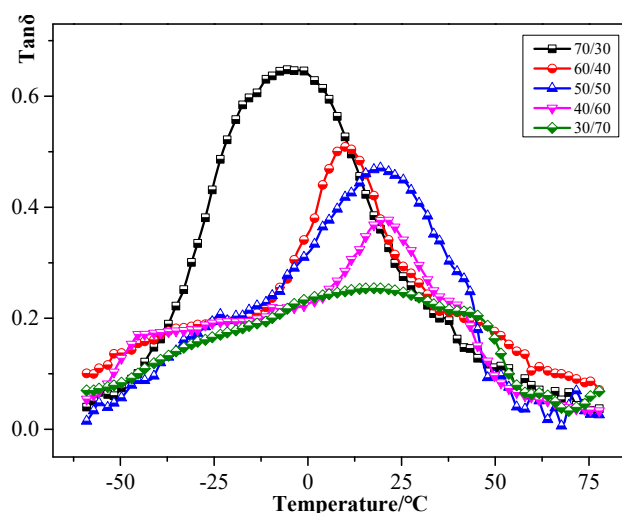


Figure 5. Dynamic mechanical properties of PU/EP grafted blends.

Table 4. Dynamic mechanical data of PU/EP grafted blends.

PU/EP.	Tg (°C)	Max (Tanδ)
70/30	−5.1	0.665
60/40	10.9	0.503
50/50	20.7	0.474
40/60	20.9	0.392
30/70	21.0	0.256

4.6. Comparison of the Antifouling Effect Shown in the Marine Hanging Plate Test

In this section, an performance verification test of the antifouling effect of the antifouling coating prepared with hydrophilic modified PU/EP film-forming resin is described. The formulations of three antifouling coating templates are roughly presented in Table 5, including a blank template with only anticorrosion coating, a hydrophilic modified antifouling coating, and a commercial antifouling coating. As shown in the table, Cu₂O, Fe₂O₃, air SiO₂, and KH550 were the components added to the antifouling agent, and these are not detailed in this paper [5,24]. Table 6 demonstrates that the template without antifouling coating was strongly attached to marine fouling organisms, while the surfaces of the two antifouling coatings soaked in seawater for 60 days were in good condition, except for containing some sea mud. After immersion for 120 days, the surface state of hydrophilic modified antifouling coating was significantly better than that of commercial antifouling coating. Bubbles appeared on the surface of the latter, and a small number of fouling organisms began to adhere to the edge of the template. To summarize, the antifouling effect of hydrophilic modified antifouling coatings was found to be better than that of commercial antifouling coatings under experimental conditions. However, the hydrophilic antifouling coatings needs to be explored further due to the influences of the production cost and complexity of antifouling coating performance testing.

Table 5. The template formulations of the three comparative antifouling coatings.

Sample	Anticorrosive Paint	Commercial Marine Antifouling Paint	Amount (phr)				
			PU/EP Grafted Blends	Cu ₂ O	Fe ₂ O ₃	Air SiO ₂	KH550
Blank sample	✓	/	/	/	/	/	/
Hydrophilic sample	✓	/	100	20	10	4	2
Unmodified	✓	✓	/	/	/	/	/

Table 6. Comparison of the antifouling effect shown in the marine hanging plate test.

Days	Blank Sample	Hydrophilic Modified	Unmodified
0			
60			
120			

5. Conclusions

Through hydrophilic modification of film-forming resin for antifouling coatings, PU/EP grafted blends were designed and synthesized by the two-step method presented in this paper. Additionally, the target products were verified by infrared spectroscopy. As expected, the test results demonstrated that the hydrophilic modification of film-forming resin has no significant effect on the mechanical properties. The specific conclusions drawn were as follows.

(1) With an increase in the PU content, the hydrophilic properties of PU/EP grafted blends significantly improved. It can be concluded from the results for the hydrophilic and mechanical properties of the materials that the comprehensive properties of PU/EP grafted blends are the best when the PU content is close to 50%. Thus, the introduction of hydrophilic segments into polymers is a feasible way to improve the hydrophilicity of materials.

(2) According to the dynamic mechanical properties of PU/EP grafted blends, the compatibility of PU/EP grafted blends worsens when the content of PU is less than 40%. When polymer blending modifications are performed, the phase compatibility between them should be considered. Therefore, the content of PU should be 40–50% when PU is used for hydrophilic modification of EP material.

(3) The antifouling effect of the hydrophilic modified antifouling coating is better than that of the commercial antifouling coating under experimental conditions.

In future studies on antifouling coatings, the hydrophilicity of film-forming resin could be improved to promote the antifouling effect of antifouling coatings. However, the

experimental state of the hanging plate in this paper was static. The antifouling effect of antifouling coatings under dynamic conditions could be explored as a next step.

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Data Availability Statement: The data presented in this study are available on request from the corresponding author.

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